

SI Session 2 Review

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1 Problem 1

$$\frac{dy}{dx} = \frac{x \cos(x^2)}{\sin(y)}, \quad y(0) = \frac{\pi}{2}$$

1.1 Solution

$$\int \sin(y) dy = \int x \cos(x^2) dx$$

$$-\cos(y) = \frac{\sin(x^2)}{2} + K_1$$

Find K_1

$$-\cos(\pi/2) = \frac{\sin(0)}{2} + K_1$$

$$K_1 = 0$$

Solve for y

$$\cos^{-1}\left(-\frac{\sin(x^2)}{2}\right) = y$$

2 Problem 2

$$\frac{dy}{dx} = y^2 \frac{1 - x^2}{x(1 + x^2)}$$

2.1 Solution

$$\int \frac{1}{y^2} dy = \int \frac{1 - x^2}{x(1 + x^2)}$$

Split the integrals

$$\int \frac{1}{y^2} dy = \frac{-1}{y}$$

$$\begin{aligned}\int \frac{1-x^2}{x(1+x^2)} dx &= \int \frac{A}{x} + \frac{Bx+C}{x^2+1} dx \\ A=1, \quad B=-2, \quad C=0 \\ \int \frac{1}{x} + \frac{-2x}{x^2+1} dx &= \ln \left| \frac{x}{1+x^2} \right| \\ \frac{-1}{y} &= \ln \left| \frac{x}{x^2+1} \right| + K_1 \\ y &= (\ln \left| \frac{x^2+1}{x} \right| + K_2)^{-1}\end{aligned}$$

3 Problem 3

$$\frac{dy}{dx} = \frac{y+1}{y} 2xe^{x^2}$$

3.1 Solution

$$\int \frac{y}{y+1} dy = \int 2xe^{x^2} dx$$

Split the integrals

$$\begin{aligned}\int \frac{y+1-1}{y+1} dy &= \int 1 - \frac{1}{y+1} dy = y - \ln |y+1| \\ \int 2xe^{x^2} dx &= e^{x^2} \\ y - \ln |y+1| &= e^{x^2} + K_1\end{aligned}$$

4 Problem 4

$$\frac{dy}{dx} = \frac{x+y}{x-y}$$

4.1 Solution

$$\begin{aligned}\frac{dv}{dx}x + v &= \frac{x+xv}{x-xv} \\ \frac{dv}{dx}x &= \frac{v^2+1}{1-v} \\ \int \frac{1}{v^2+1} - \frac{v}{v^2+1} dv &= \int \frac{1}{x} dx \\ \tan^{-1}(v) - \frac{1}{2} \ln |v^2+1| &= \ln |x| + K_1 \\ \tan^{-1}\left(\frac{y}{x}\right) - \ln \left| \frac{(y^2+x^2)^{1/2}}{x} \right| &= \ln |x| + K_1\end{aligned}$$

5 Problem 5

$$\frac{dy}{dx} = \frac{x + 2y}{2x - y}$$

5.1 Solution

$$\frac{dv}{dx}x + v = \frac{x + 2xv}{2x - xv}$$

$$\frac{dv}{dx}x + v = \frac{1 + 2v}{2 - v}$$

$$\frac{dv}{dx}x = \frac{v^2 + 1}{2 - v}$$

$$\int \frac{2}{v^2 + 1} - \frac{v}{v^2 + 1} dv = \int \frac{1}{x} dx$$

$$2 \tan^{-1}(v) - \frac{1}{2} \ln |v^2 + 2| = \ln |x| + K_1$$

$$2 \tan^{-1}\left(\frac{y}{x}\right) - \ln \left| \frac{(y^2 + x^2)^{1/2}}{x} \right| = \ln |x| + K_1$$

6 Problem 6

$$\frac{dy}{dx} = \frac{x^2 + 3xy}{x^2 - xy}$$

6.1 Solution

$$\frac{dv}{dx}x + v = \frac{x^2 + 3x^2v}{x^2 - x^2v}$$

$$\frac{dv}{dx}x = \frac{v^2 + 2v + 1}{1 - v}$$

$$\int \frac{1 - v}{(1 + v)^2} dv = \int \frac{1}{x} dx$$

Split the integral

$$\int \frac{1 - v}{(1 + v)^2} dv = \int \frac{A}{1 + v} + \frac{B}{(1 + v)^2} dv$$

$$A = -1, \quad B = 2$$

$$\int \frac{-1}{v + 1} + \frac{2}{(v + 1)^2} dv = -\ln |1 + v| - \frac{2}{v + 1}$$

$$\int \frac{1}{x} dx = \ln |x|$$

$$-\ln |1+v| - \frac{2}{v+1} = \ln |x| + K_1$$

$$-\ln \left| \frac{x+y}{x} \right| - \frac{2x}{x+y} = \ln |x| + K_1$$

7 Problem 7

$$(ye^{xy} + 2x)dx + (xe^{xy} + 3y^2)dy = 0$$

7.1 Solution

Test for exactness

$$\frac{\partial M}{\partial y} = e^{xy} + xye^{xy} + 0$$

$$\frac{\partial N}{\partial x} = e^{xy} + xye^{xy} + 0$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\int M(x,y)dx = \int (ye^{xy} + 2x)dx$$

$$F = e^{xy} + x^2 + h(y)$$

$$\frac{\partial F}{\partial y} = xe^{xy} + 0 + h'(y)$$

Now set $\partial F_y = N(x,y)$

$$h'(y) = 3y^2$$

$$\int h'(y)dy = \int 3y^2 dy$$

$$h(y) = y^3 + K_1$$

Plug into F

$$K_2 = e^{xy} + x^2 + y^3$$

8 Problem 8

$$(2xy + y^2 + \sin(x))dx + (x^2 + 2xy - \cos(y))dy = 0$$

8.1 Solution

Test for exactness

$$\frac{\partial M}{\partial y} = 2x + 2y + 0$$

$$\frac{\partial N}{\partial x} = 2x + 2y - 0$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\int M(x, y)dx = \int 2xy + y^2 + \sin(x)dx$$

$$F = x^2y + xy^2 - \cos(x) + h(y)$$

$$\frac{\partial F}{\partial y} = x^2 + 2xy + 0 + h'(y)$$

Now set $\partial F_y = N(x, y)$

$$h'(y) = -\cos(y)$$

$$\int h'(y)dy = \int -\cos(y)dy$$

$$h(y) = -\sin(y) + K_1$$

Plug into F

$$K_2 = x^2y + xy^2 - \cos(x) - \sin(y)$$

9 Problem 9

$$(3x^2y + \sin(y))dx + (x^3 + x \cos(y))dy = 0$$

9.1 Solution

Test for exactness

$$\frac{\partial M}{\partial y} = 3x^2 + \cos(y)$$

$$\frac{\partial N}{\partial x} = 3x^2 + \cos(y)$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\int M(x, y)dx = \int 3x^2y + \sin(y)dx$$

$$F = x^3y + x \cos(y) + h(y)$$

$$\frac{\partial F}{\partial y} = x^3 + x \cos(x) + h'(y)$$

Now set $\partial F_y = N(x, y)$

$$h'(y) = 0$$

$$\int h'(y) dy = 0$$

$$h(y) = K_1$$

Plug into F

$$K_2 = x^3 y + x \sin(y)$$

10 Problem 10

$$\frac{dy}{dx} - \tan(x)y = \sec(x)$$

10.1 Solution

Find the integrating factor

$$IF = e^{\int \tan(x) dx} = \cos(x)$$

$$\int d(\cos(x)y) = \int (1) dx$$

$$y = x \sec(x) + K_1 \sec(x)$$

11 Problem 11

$$\frac{dy}{dx} + \frac{x}{x^2 + 1}y = \frac{x}{x^2 + 1}$$

11.1 Solution

Find the integrating factor

$$IF = e^{\int \frac{x}{x^2+1}} = (x^2 + 1)^{1/2}$$

$$\int d((x^2 + 1)^{1/2}y) = \int \frac{x}{(x^2 + 1)^{1/2}} dx$$

$$y = 1 + K - 1(x^2 + 1)^{-1/2}$$

12 Problem 12

$$\frac{dy}{dx} + \sec(x)y = \tan(x)$$

12.1 Solution

Find the integrating factor

$$IF = e^{\int \sec(x) dx} = \sec(x) + \tan(x)$$

$$\int d((\sec(x) + \tan(x))y) = \int \tan(x)(\sec(x) + \tan(x))dx$$

Split the integral

$$\int \frac{\sin(x)}{\cos^2(x)} dx + \int \frac{\sin^2(x)}{\cos^2(x)} dx$$
$$y(\tan(x) + \sec(x)) = \sec(x) + \tan(x) - x + K_1$$